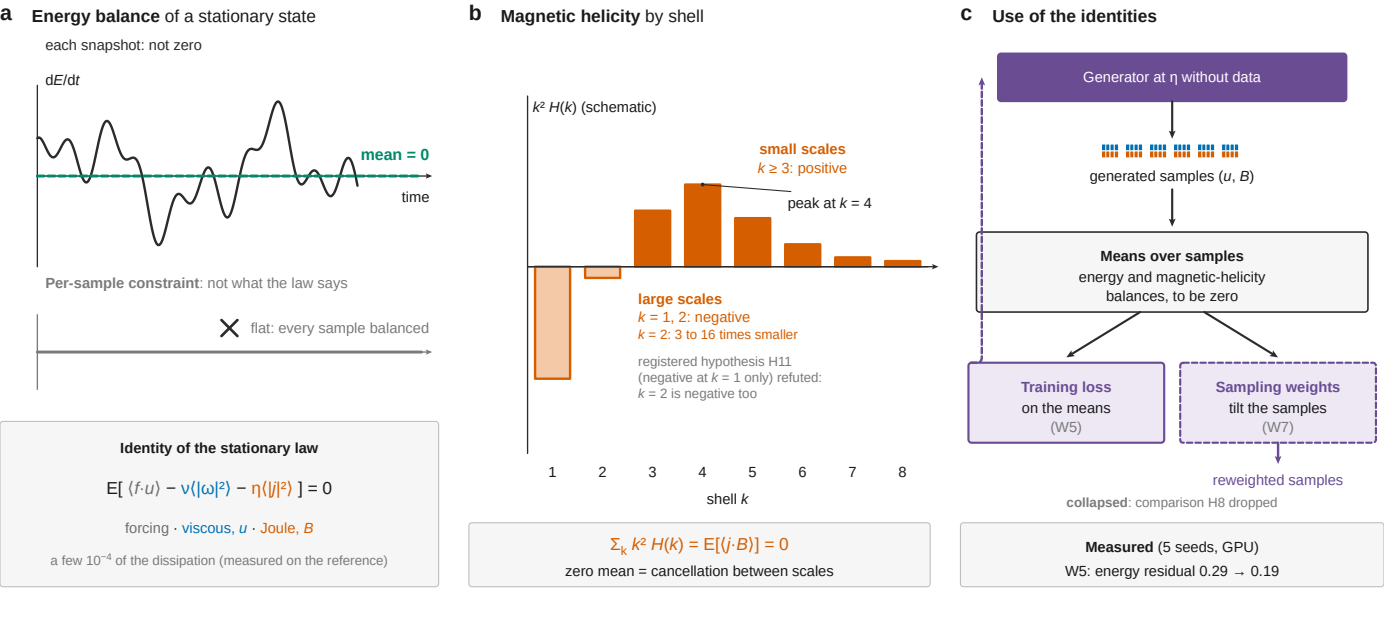

a
Energy balance of a stationary state
each snapshot: not zero
dE/dt
mean = 0
time
Per-sample constraint: not what the law says
flat: every sample balanced
Identity of the stationary law
E[ ⟨f·u⟩ − ν⟨|ω|²⟩ − η⟨|j|²⟩ ] = 0
forcing · viscous, u · Joule, B
a few 10⁻⁴ of the dissipation (measured on the reference)
b
Magnetic helicity by shell
k² H(k) (schematic)
small scales
k ≥ 3: positive
peak at k = 4
large scales
k = 1, 2: negative
k = 2: 3 to 16 times smaller
registered hypothesis H11
(negative at k = 1 only) refuted:
k = 2 is negative too
1 2 3 4 5 6 7 8
shell k
Σ_k k² H(k) = E[⟨j·B⟩] = 0
zero mean = cancellation between scales
c
Use of the identities
Generator at η without data
generated samples (u, B)
Means over samples
energy and magnetic-helicity
balances, to be zero
Training loss
on the means
(W5)
Sampling weights
tilt the samples
(W7)
reweighted samples
collapsed: comparison H8 dropped
Measured (5 seeds, GPU)
W5: energy residual 0.29 → 0.19